

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

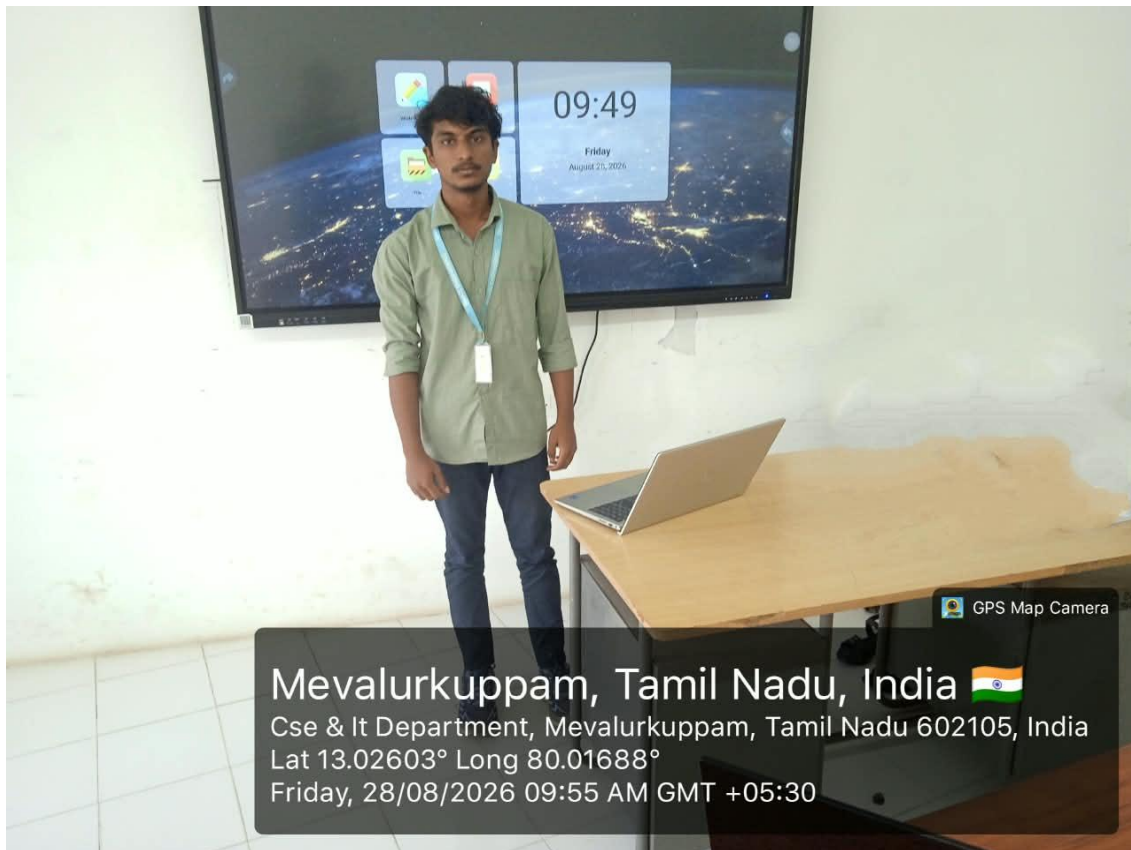
5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Reactive Agent-Based Traffic Control System



1. Introduction

- A Reactive Agent-Based Traffic Control System is an intelligent traffic management system that observes current traffic conditions and immediately responds with suitable actions.
- Unlike traditional traffic signals that use fixed timings, reactive agents continuously monitor the traffic environment.
- The system can respond to:
 - Traffic congestion
 - Accidents
 - Emergency vehicles
 - Pedestrian movement
 - Changes in vehicle density
- The main objectives are:

- Reduce traffic congestion
- Minimize vehicle waiting time
- Improve road safety
- Maintain smooth and efficient traffic flow

2. What is a Reactive Agent?

A reactive agent is an intelligent agent that:

- Observes the current environment.
- Receives information through sensors.
- Processes the current traffic condition.
- Applies predefined decision rules.
- Takes an immediate action based on the observed condition.
- Does not necessarily maintain a detailed history of previous situations.

Example Rule:

IF traffic density is high on Road A
THEN increase the green-light duration for Road A.

3. System Components

The major components of the system are:

- Sensors
- Reactive Agent
- Traffic Signal Controller
- Communication Network

Sensors

The sensors collect real-time information from the traffic environment.

- CCTV cameras
- Vehicle detectors
- Infrared sensors
- GPS information
- Emergency vehicle detection systems

Reactive Agent

The agent analyzes the information received from the sensors and makes immediate decisions.

- Receives real-time traffic information.
- Determines the current traffic condition.
- Identifies congestion and unusual traffic situations.
- Applies predefined decision rules.
- Determines appropriate signal changes.
- Sends commands to the traffic signal controller.

Traffic Signal Controller

The traffic signal controller executes the decisions made by the reactive agent.

- Controls red, yellow, and green traffic signals.
- Changes signal timing according to traffic conditions.
- Increases green-light duration for roads with higher traffic density.
- Gives priority to emergency vehicles when detected.
- Helps reduce unnecessary waiting time.

Communication Network

The communication network enables information exchange between different components.

- Transfers traffic information between intersections.
- Connects traffic sensors with the reactive agent.
- Transfers control commands to traffic signals.
- Enables coordination between traffic-control agents at nearby intersections.
- Supports real-time traffic management.